Biofilm Growth and Stress on a Real Molar

3D demonstrations of the growth model, with converged stresses and no experimental data

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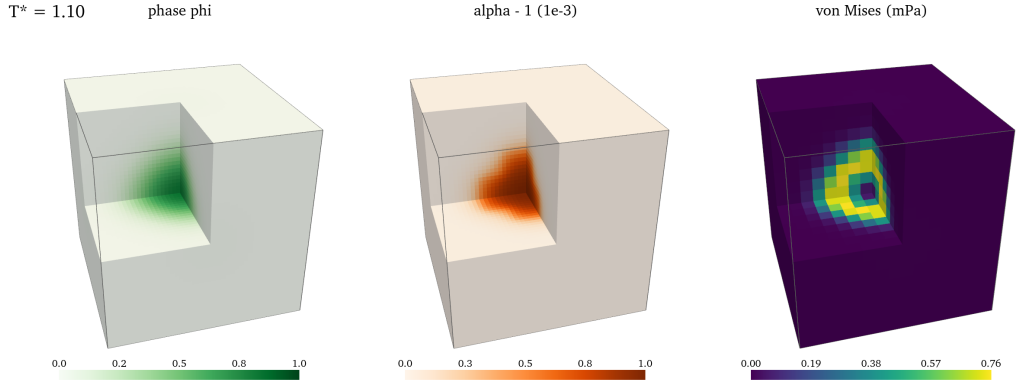
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20 October 2026

What I show

- Growth model of Klempt et al. (2024): phase ϕ with $\dot{\phi} = \beta \nabla^2 \phi + k_\alpha \alpha$ (plus the bound on ϕ), growth $\mathbf{F}_g = \alpha \mathbf{I}$, stiffness $E(\phi^2 + f)$, $E = 10\,\mathrm{Pa}$, $\nu = 0.49$.
- Added in this work: a nutrient field supplied by the saliva and used up by the biofilm (Klempt et al. 2024, Eq. 35), and a growth limit by pressure, $\omega = \min\big(1, \max(0, 1 - p/p_h)\big)$.
- Geometry: the thesis's 2 mm cube for verification, then a real lower first molar at its real size (surface from a public CC BY 4.0 data set).
- Every stress I quote is checked against mesh refinement; values that do not converge are not quoted.
- Only the verification problem is a thesis result. **No experimental data are used.**

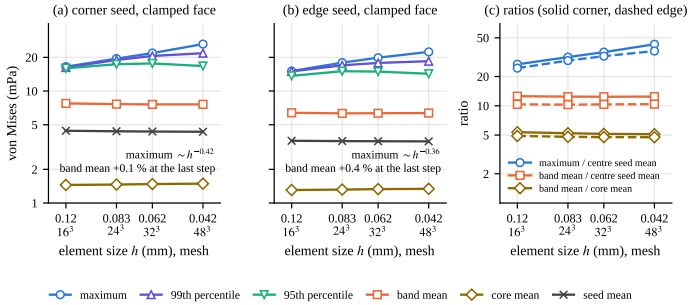
1. Verification: the thesis problem in 3D



$\beta = 0.02\,\mathrm{mm}^2/T^*$, $k_\alpha = 10^{-3}/T^*$, $E = 10\,\mathrm{Pa}$, $\nu = 0.49$, $T^* = 1.1$; one octant cut away. Field on 48^3 (reference solution), stress on 16^3.

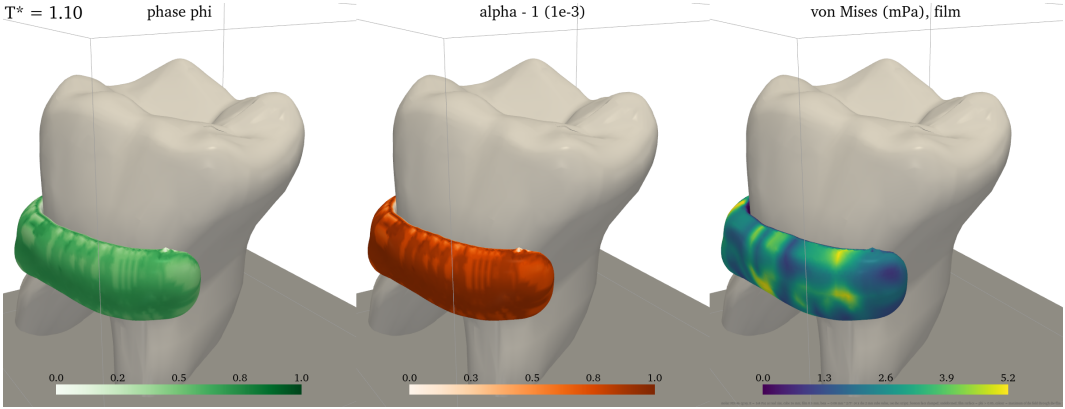
Seed von Mises at $T^* = 1.1$: 6.19×10^{-4} Pa (thesis, 16^3: 6.18×10^{-4} Pa); largest value 0.79 mPa, converged from 24^3 on. The stress forms a ring at the edge of the seed. ▶ video: `assets/anim_wp3_seed_3d.mp4`

2. Stress next to a fixed surface: what converges



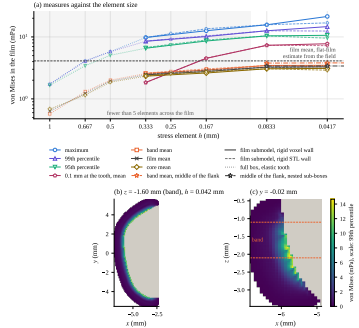
Seed on a clamped face of the cube, meshes 16^3 to 48^3. The mean stress in the biofilm next to the face converges (less than 1 % change from 24^3 on) and is 10 to 12 times the seed stress of the verification problem. The pointwise maximum, where the clamped face meets a free wall, grows as about $h^{-0.4}$: a corner of the elastic problem. I quote means and percentiles only.

3. A biofilm on a real molar



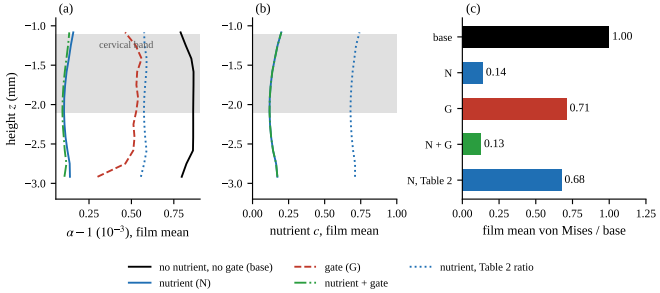
Lower first molar at real size (crown about 10×12 mm) in a 16 mm box, root clamped. A 0.5 mm film on one flank at the cervical third of the crown. β is scaled to $0.08\,\mathrm{mm}^2/T^*$ so that the film spreads by the same share of its thickness as the thesis seed (an assumption). Growth field converged to 2 % (48^3 to 192^3). ▶ video: `assets/anim_real_tooth_3d.mp4`

4. Resolving the stress in a 0.5 mm film



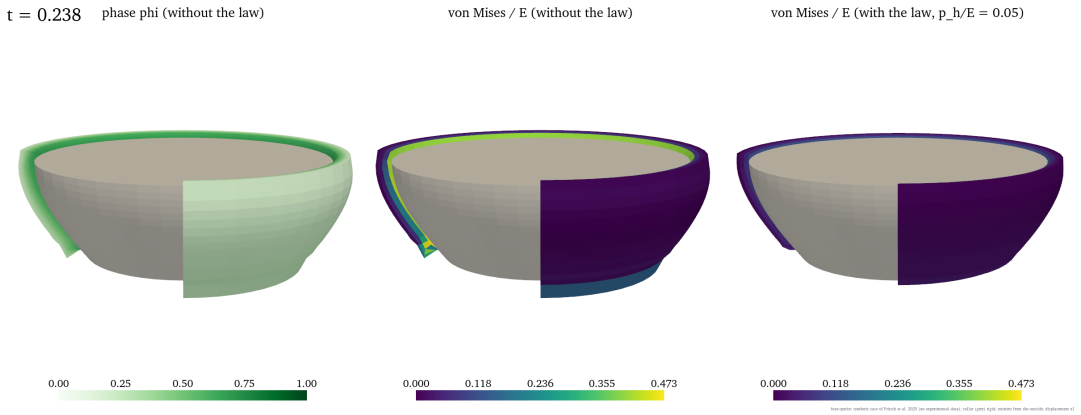
On meshes of the whole box the film is one or two elements thick and the stress does not converge. I solve the film alone with the tooth as rigid (1000 times stiffer; 0.1 % effect), down to $h = 0.028$ mm. With six elements across the film the means converge: film **3.3 mPa** (± 1 %), cervical band 3.5, the 0.1 mm next to the tooth 7.8, 95th percentile 10.5 mPa. The maximum sits at a corner of the tooth wall and has no converged value.

5. Nutrient and the growth limit on the molar



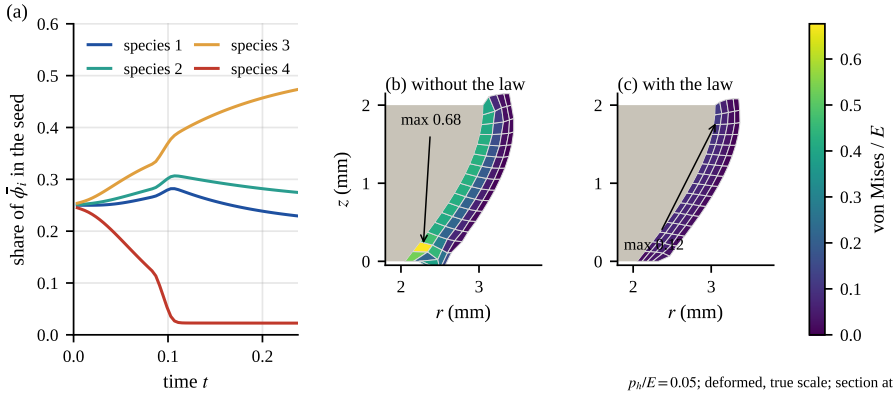
Nutrient from the saliva, consumption $g\phi c$ with $g/d = 10\,\mathrm{mm}^{-2}$ (chosen; 10 times the Table 2 ratio): c falls to 0.07 at the tooth, and growth and stress drop to about one seventh. At the Table 2 ratio ($c = 0.62$ to 0.78) they drop to 0.68. The growth limit alone (p_h half the film pressure without it, a choice) gives 0.71; with the nutrient, the nutrient dominates. Stress: end states re-solved with the film model. ▶ video: `assets/anim_real_tooth_nutrient_gate_3d.mp4`

6. Four species on a tooth-like collar (synthetic data)



A 0.25 mm biofilm layer on a crown-shaped collar (radius 2.05 to 3.05 mm, 2 mm high), inner face bonded, nutrient from the outside. Point model: the four-species case of Fritsch et al. (2025) with their true parameters (synthetic, no experimental data). One quarter is cut away. Growth is large here (displacement up to 0.31 mm), drawn at true scale. ▶ video: `assets/anim_cervical_f4_3d.mp4`

6. Section and composition



(a) Species 3 takes over the seed (share 0.47 at the end), species 4 almost disappears (0.02). (b), (c) Section at 45°: without the law the von Mises stress reaches $0.68E$ at the bonded base; with $p_h/E = 0.05$ the largest value is $0.12E$.

Summary

- The 3D view reproduces the thesis value of the seed stress (0.2 % difference).
- Next to a fixed surface the biofilm stress is 10 to 12 times the free-seed value (converged band mean); pointwise maxima at corners have no converged value.
- On a real molar, a 0.5 mm film needs about six elements across it; then the film mean is 3.3 mPa ($E = 10$ Pa), highest in the 0.1 mm next to the tooth (2.4 times the mean).
- A nutrient limited by consumption lowers growth and stress more than the pressure limit; the pressure limit removes stress concentrations in the cube and on the collar.
- Next at Keio: the calibrated parameters of Chapter 3 on this molar, the gate with the film stress in every step, and an implant.

Appendix A: notation and equations

ϕ	biofilm phase (volume fraction)
α	growth, $\mathbf{F}_g = \alpha\mathbf{I}$, $\alpha(0) = 1$; growth is $\alpha - 1$
β, k_α	diffusion of ϕ, growth rate (Klempt et al. 2024, Eq. 34/36)
$E(\phi^2 + f)$	stiffness, $f = 10^{-3}$ (void)
p, p_h, ω	pressure $-\frac{1}{3}\,\mathrm{tr}\,\boldsymbol{\sigma}$, homeostatic pressure, growth limit
c, d, g	nutrient, its diffusion, consumption rate (Klempt et al. 2024, Eq. 35)
$\phi_i, \psi_i, \bar{\phi}_i = \phi_i\psi_i$	point model (Klempt et al. 2026): volume fraction, living fraction
c^*	nutrient scale of the point model (25 in the four-species case)

Cube and molar: $\dot{\phi} = \beta\nabla^2\phi + k_\alpha\alpha - P\,[\max(0, \phi - 1) + \min(0, \phi)]$, $P = 5$; $\dot{\alpha} = k_\alpha\phi\,c\,\omega$ ($c = 1$ without the nutrient, $\omega = 1$ without the limit).
Collar: growth from the point model in the element of this work, scaled by the same ω.

Appendix B: sources

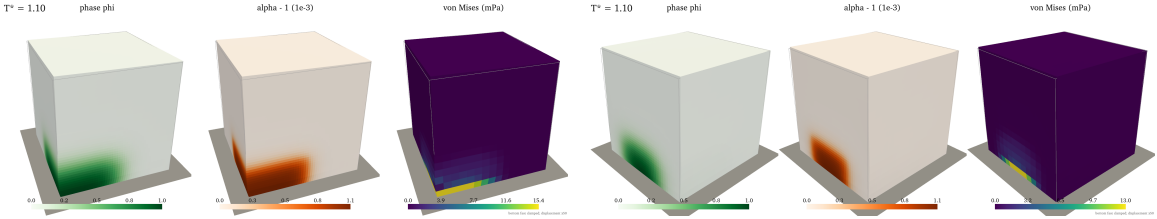
- Tooth surface: C. Diaz, C. A. Ferro Sánchez, O. Campo, *Data of synthetic 3D models of the human jaw, including teeth, ligaments, and bone structures*, Mendeley Data V1 (2024), doi:10.17632/xjsx7nfhj8.1, licence CC BY 4.0; file `D46.stl` (lower right first molar), used unchanged.
- Growth model and parameters: F. Klempt et al., Biomech. Model. Mechanobiol. (2024).
- Four-species synthetic case: L. Fritsch et al., arXiv:2512.15145 (2025), Case II, true parameters.
- Scripts and reports: `ansys_usermat/apdl/` (`real_tooth_*.py`, `results/2026-10-3d/`).

Appendix C: settings

case	geometry	mesh (field / stress)	values
verification	2 mm cube, seed in the centre	48^3 / 16^3	thesis (Ch. 4)
wall study	cube, seed on a clamped face	96^3 / 16^3–48^3	thesis values
real molar	16 mm box, film 0.5 mm	96^3, 192^3 / film model to 0.028 mm	β scaled $\times 4$
nutrient, limit	real molar	96^3 / film model 0.083 mm	g/d and p_h chosen
collar	quarter ring, $4 \times 12 \times 16$	element of this work	Fritsch et al. 2025

Small strain, hex8 with B-bar in the cube and on the molar (rigid tooth, $E_{\mathrm{tooth}} = 10^4$ Pa where elastic); finite strain on the collar. Growth with the nutrient: $\dot{\alpha} = k_\alpha \phi c \, \omega$ (the factor c is an assumption; Eq. 36 as printed has none). Nutrient quasi-static, $0 = d\nabla^2 c - g\phi c$, $c = 1$ at the saliva.

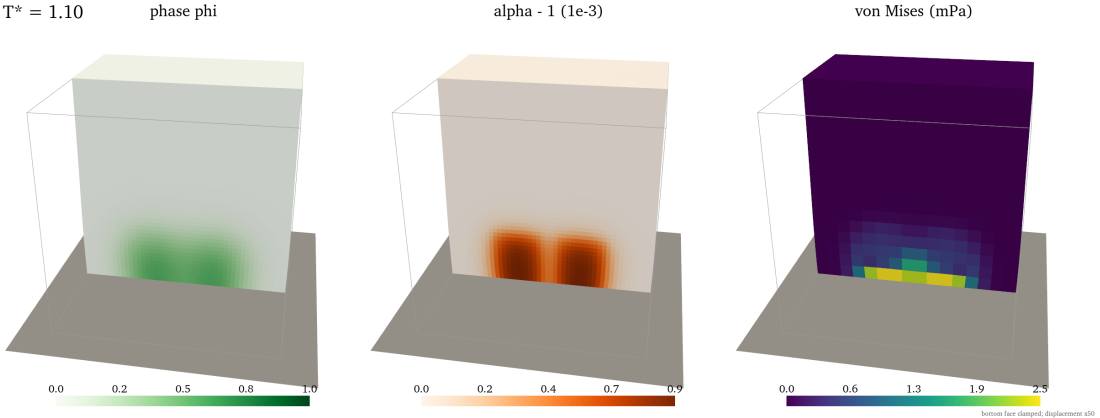
Backup: seed in a corner and on an edge (cube)



Same model and seed volume as the verification problem; bottom face clamped (the substrate). Left: corner (three faces), right: edge (two faces). The biofilm spreads along the walls; the stress concentrates in the biofilm next to the clamped face: the mean over its bottom quarter millimetre is 7.6 mPa (corner) and 6.4 mPa (edge), 10 to 12 times the seed stress of the centre case (0.61 mPa) and about 5 times the mean of the layer above it; these means change by less than 1 % from 24^3 to 48^3. The pointwise maximum sits where the clamped face meets a free wall and grows with refinement (16 to 26 mPa, about $h^{-0.4}$), so the colour scale stops at the 95th percentile, about 15 mPa, which does not depend on the mesh.

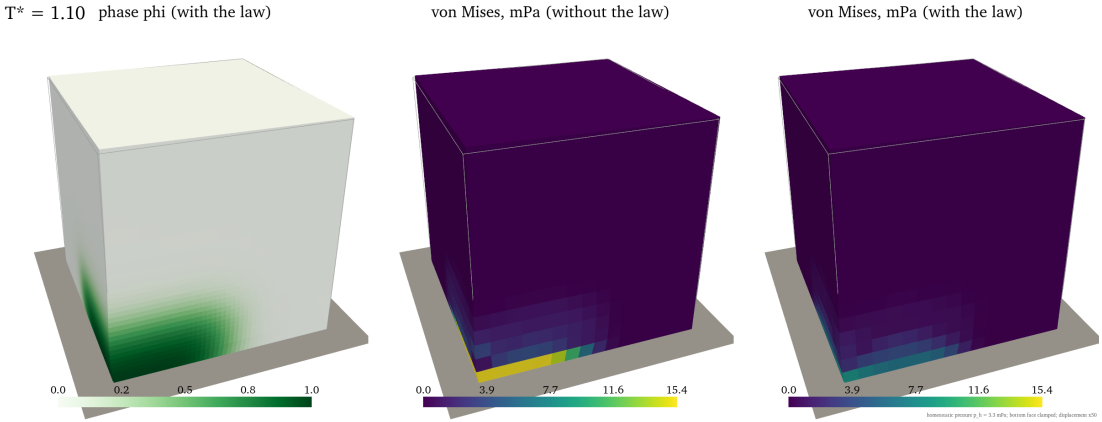
▶ video: `assets/anim_seed_corner_3d.mp4` ▶ video: `assets/anim_seed_edge_3d.mp4`

Backup: two seeds that grow into each other (cube)



Two seeds of 0.5 mm on the clamped bottom face, 0.125 mm apart; front half cut away; same model as the verification problem. The two seeds join, and the stress is highest along the clamped face under both of them. ▶ video: `assets/anim_two_3d.mp4`

Backup: the growth limit in the cube



Corner seed, same colour scale in both stress panels. Without the law the stress in the biofilm next to the clamped face reaches about 15 mPa (95th percentile, the top of the colour scale); with the law it stays well below. $p_h = 3.3$ mPa is chosen by me (0.4 times the 95th percentile of the pressure without the law), not taken from a paper. ▶ video: `assets/anim_law_3d.mp4`